Silicon Nitride Photonic Integrated Devices for Navigation Systems

Rifat Nazneen



Department of Electrical and Computer Engineering
McGill University
Montréal, Québec, Canada

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Abstract

Integrating photonic devices into commercial and consumer products has expanded their applications beyond research laboratories. This transition is driven by their compact size, robustness under environmental conditions such as vibrations and temperature variations, and low power consumption. Integrated photonic devices are now expanding their development in navigation technologies by providing accurate, reliable positioning, navigation, and timekeeping (PNT) sensing. Silicon nitride (SiN) photonics has emerged as a leading material due to its low loss, moderate Kerr nonlinearity coefficient, negligible two-photon absorption, and broad optical transparency. These characteristics make SiN an ideal choice for advancing integrated photonic devices, enabling high-accuracy distance and rotational movement measurements for advanced navigation systems. This thesis introduces two SiN-based photonic devices exploiting nonlinear and linear optical phenomena. The thesis discusses the design methodology, experimental validation, and design challenges encountered, along with proposed solutions. A comprehensive study on SiN-based photonic devices for navigation sensor systems is also included.

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In the first part of the thesis, we design, simulate, and experimentally characterize a 400 µm SiN microring racetrack resonator with a 30 µm coupling length. The waveguide geometry is optimized for the normal dispersion of -359 ps/nm.km at 1547 nm wavelength. We experimentally characterized the microring resonator, determining a loaded quality factor (Q_L) of 77,800 at 1546.19 nm, with an on-chip insertion loss of 0.3 dB. We compare two measured 400 µm radius microring resonators: one in a racetrack configuration and the other a circular ring without a coupling length. The circular ring exhibits a higher insertion loss of 1.5 dB compared to the racetrack microring resonator due to the curvature waveguide in the coupling region. This comparison validates the advantage of the racetrack resonator design for nonlinear optics. In numerical simulations, we demonstrate the resonator's potential to generate six comb lines with a 0.48 nm spacing using two 60 mW pump signals through non-degenerate four-wave mixing (FWM). These comb lines enable precise positioning and timekeeping applications using a LIDAR and optical clock. The second part of the thesis focuses on the development of an efficient double lead zirconate titanate (PZT) actuator-based SiN phase modulator for optical gyroscope applications using piezoelectric and stress-optic effects. By optimizing key geometric parameters, the half-wave voltage (V_π) is significantly reduced from 65 V to 9.1 V. The optimized design adjusts the distance between two PZT structures from 22.8 µm to 2 µm, reducing the top cladding from 8 µm to 2 µm and increasing the waveguide width from 2.8 µm to 3.7 µm. The maximum tensile stress that mechanically propagates from the PZT structure to the

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waveguide structure is 1×10^8 N/m^2 in the core to achieve a π-shift at low voltage. The optimized design reduces its width by 10 %.

The development of two SiN-integrated photonic devices discussed in this thesis is a key contribution to improving a navigation system's (GNSS) performance through PNT sensors. The integration of an optical frequency comb and an optical gyroscope provides precise distance measurement, timekeeping, and rotational movements of a receiver, potentially enhancing the overall accuracy and reliability of navigation systems.

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Abrégé

L'intégration des dispositifs photoniques dans les produits commerciaux et grand public a élargi leurs applications au-delà des laboratoires de recherche. Cette transition est rendue possible par leur taille compacte, leur robustesse face aux vibrations et aux variations de température, ainsi que leur faible consommation d'énergie. Les dispositifs photoniques intégrés connaissent aujourd'hui un essor dans les technologies de navigation en fournissant des mesures précises et fiables de positionnement, de navigation et de synchronisation (PNT). La photonique à base de nitrure de silicium (SiN) s'impose comme matériau de référence grâce à ses faibles pertes optiques, son coefficient de non-linéarité de Kerr modéré, son absorption biphotonique négligeable et sa large transparence optique. Ces propriétés font du SiN un excellent choix pour le développement de dispositifs photoniques intégrés destinés à des mesures de distance et de mouvements rotationnels de haute précision pour les systèmes de navigation avancés. Cette thèse présente deux dispositifs photoniques en SiN exploitant des phénomènes optiques linéaires et non linéaires. Elle décrit la méthodologie de conception, la validation expérimentale, les défis techniques

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rencontrés et les solutions proposées. Une étude approfondie sur les dispositifs photoniques à base de SiN pour les systèmes de capteurs de navigation est également incluse.

Dans la première partie, nous concevons, simulons et caractérisons expérimentalement un résonateur optique à micro-anneau de type racetrack en SiN de 400 µm, avec une longueur de couplage de 30 µm. La géométrie du guide d'onde est optimisée pour une dispersion normale de -359 ps/nm · km à la longueur d'onde de 1547 nm. Le résonateur est caractérisé expérimentalement avec un facteur de qualité chargé (Q_L) de 77,800 à 1546,19 nm et une perte d'insertion sur puce de 0,3 dB. Nous comparons deux résonateurs optiques à micro-anneau de 400 µm de rayon: l'un en configuration racetrack, l'autre sous forme annulaire circulaire sans longueur de couplage. Le résonateur circulaire présente une perte d'insertion plus élevée de 1,5 dB en raison de la courbure dans la zone de couplage, ce qui confirme l'avantage de la configuration racetrack pour les applications en optique non linéaire. Des simulations numériques montrent la génération de six lignes de peigne optique espacées de 0,48 nm à l'aide de deux pompes de 60 mW, via un mélange à quatre ondes non dégénéré. Ces lignes de peigne permettent des applications de positionnement et de synchronisation de haute précision utilisant un système LIDAR et une horloge optique.

La seconde partie de la thèse porte sur le développement d'un modulateur de phase SiN basé sur des actionneurs doubles PZT (plomb-zirconate-titanate) pour des applications de gyroscope optique, utilisant les effets piézoélectrique et optique de contrainte. En optimisant certains paramètres géométriques clés, la tension de demi-onde (V_π) est significativement

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réduite de 65 V à 9,1 V. La conception optimisée réduit la distance entre les deux structures PZT de 22,8 µm à 2 µm, diminue l'épaisseur du revêtement supérieur de 8 µm à 2 µm, et augmente la largeur du guide d'onde de 2,8 µm à 3,7 µn. La contrainte de traction maximale transmise mécaniquement de la structure PZT vers le guide d'onde atteint 1×10^8 N/m^2 dans le cœur pour générer un déphasage de π à basse tension. Cette conception permet aussi une réduction de la largeur de 10 %.

Le développement des deux dispositifs photoniques intégrés en SiN présenté dans cette thèse constitue une contribution essentielle à l'amélioration des performances des systèmes de navigation (GNSS) via les capteurs PNT. L'intégration d'un peigne de fréquences optiques et d'un gyroscope optique permet des mesures précises de distance, de synchronisation temporelle et de mouvements de rotation d'un récepteur, améliorant ainsi la précision et la fiabilité globales des systèmes de navigation.

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List of Acronyms

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ADS	Asymmetric Double Stripe.
CMOS	Complementary Metal–Oxide–Semiconductor.
CW	Continuous Wave.
DC	Directional Coupler (also Direct Current).
EDFA	Erbium-Doped Fiber Amplifier.
ER	Extinction Ratio.
FSR	Free-Spectral Range.
FWHM	Full Width Half Maximum.
FWM	Four-Wave Mixing.
GC	Grating Coupler.
GNSS	Global Navigation Satellite System.
GPS	Global Positioning System.
GVD	Group Velocity Dispersion.
INS	Inertial Navigation System.

LIDAR Light Detection and Ranging.

MEMS Micro-Electro-Mechanical Systems.

MZI Mach Zehnder Interferometer.

NIST National Institute of Standards and Technology.

OFC Optical Frequency Comb.

OSA Optical Spectrum Analyzer.

PNT Positioning, Navigation, and Timekeeping.

Pt Platinum.

PZT Lead Zirconate Titanate.

Q-factor Quality factor.

SiN Silicon Nitride.

SiO2 Silicon Dioxide.

SiPh Silicon Photonics.

SOI Silicon-on-Insulator.

TE Transverse Electric.

TeO2 Tellurium Oxide.

TiW Titanium-Tungsten.

TPA Two-Photon Absorption.

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